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Numerical and experimental study on spatter formation mechanisms and suppression method during ring-mode laser spiral scanning welding of T2 copper

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Article

Abstract

Ring-mode laser spiral scanning welding (RMLSSW) is an innovative technology for low-spatter manufacturing of thin-walled components in new energy systems. However, the spatter-free process window of RMLSSW remains empirical. This study integrates experiments and simulations to elucidate how spatiotemporal energy distribution governs spatter initiation and suppression. The spatter formation mechanism of RMLSSW is initiated by insufficient preheating, which causes sharp absorptivity fluctuations. These fluctuations generate steep thermal gradients and transverse molten pool vortices that shear the keyhole, culminating in violent evaporation at the keyhole crown and

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CRediT authorship contribution statement

subsequent spatter ejection. Power of ring (P_r) and spiral pitch (d) are identified as the critical parameters for achieving spatter suppression. Simulations reveal that increasing P_r or decreasing d raises substrate temperature and laser absorptivity, enlarges the molten pool, stabilizes melt flow around the keyhole, and reduces the probability of spatter initiation. The enhanced recoil pressure dilates the keyhole, increasing its depth-to-diameter ratio and volume, thereby suppressing spatter induced by stochastic keyhole collapse. An optimal preheating process window of $P_r=2000\text{--}2400\text{ W}$ and $d=0.55\text{--}0.70\text{ mm}$ is proposed, within which laser–material coupling is maximized and spatter is minimized. This study provides transferable theoretical foundations and well-defined optimization criteria for achieving spatter-free, high-quality laser welding of highly reflective thin-walled materials.

Keywords

Ring-mode laser spiral scanning welding; T2 copper; Spatter formation mechanism; Spatter suppression mechanism

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Data availability

Data will be made available on request.

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Declaration of competing interest

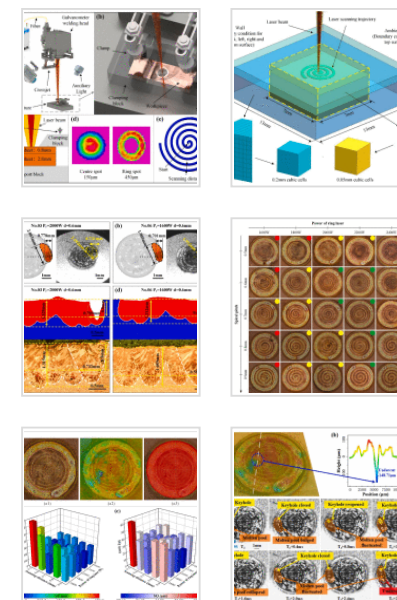
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